Candidate Seat no............

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (ME-AMT) Examination

May-2018

ME736/ME785 Machining and Forming Processes

Date: 03.05.2018, Thursday **Time: 01.30 p.m. to 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION –I

Q–1(a) **Answer following multiple choice questions. (Each carries one mark)** **[05]**

(i) Which material has very good surface hardness and hence used for table of milling machine.
(p) Cast iron (q) Steel
(r) Aluminum (s) None of the mentioned

(ii) The cutting tool wears due to-
(p) Edge wear (q) Crater wear
(r) Flank wear (s) All of the above

(iii) The built up edge in cutting tools can be eliminated by-
(p) Fast cutting speed (q) Higher rake angles
(r) High pressure cutting fluid (s) All of the above

(iv) One of the assumptions while determining shear angle during machining is 'there is no side flow of chip'.
(p) True (q) False

(v) Following is the type of machine tool vibrations -
(p) Free or Transient vibration (q) Forced vibration
(r) Self excited vibration (chatter) (s) All of these

Q-1(b) Differentiate between Machining and Forming processes. **[05]**

OR

Q-1(b) Differentiate between Orthogonal and Oblique cutting. **[05]**

Q-2 **Answer following. (Any three)** **[15]**

(i) Explain heat generation during machining at Tool-Chip Interface.
(ii) Describe the Tool-Work Thermocouple technique for measuring cutting temperature.
(iii) How abrasion wear and chemical wear of cutting tool take place? Explain in brief.
(iv) What are the benefits of minimizing vibrations of machine tool? Explain in details.

Q-3 **Answer following. (Any one)** **[10]**

(i) In an orthogonal cutting, following data have been observed-
Uncut chip thickness = 0.127 mm , Width of cut = 6.35 mm, Cutting speed = 2 m/s,
Rack angle = 10^0, Cutting force = 567 N, Thrust force = 227 N, Chip thickness = 0.228 mm.
Determine- shear angle, friction angle, shear stress along the shear plane and the power for the cutting operations. Also find the chip velocity.

(ii) Define Tool Life. How to measure a Tool Life? What are the effects of machine vibrations and impurities in work piece material on Tool Life? The Taylor tool life equation for machining C-40 steel with a 18: 4: 1 H.S.S cutting tool at a feed of 0.2 mm/min and a depth of cut of 2 mm is given by $VT^n = C$. Where n and C are constants. The following V and T observations have been noted-

V (m/min)	25	35
T (min.)	90	20

Calculate-
(1) n and C.
(2) Hence recommend the cutting speed for a desired tool life of 60 minutes.

SECTION –II

Q-4(a) Answer following multiple choice questions. (Each carries one mark) **[05]**

(i) Which of the following is not included in forming and shaping process?
(p) Rolling (q) Forging
(r) Sheet forming (s) Broaching

(ii) Depending upon the temperature the forming process can be classified as?
(p) Hot working (q) Cold working
(r) Warm working (s) All of the mentioned

(iii) In metal forming process the hardness of the material?
(p) Decreases (q) Remains same
(r) Increases then decreases (s) Increases

(iv) Which of the following method is used for analyzing metal forming processes?
(p) Slab method (q) Slip line method
(r) Both (p) & (q) (s) None of these

(v) 56 inches pipe diameter requires 56 inches width of the plate to form it into a pipe.
(p) True (q) False

Q-4(b) What is anisotropy in sheet metal? An aluminum test specimen 100 mm long, 20 mm wide and 2 mm thick is elongated to 130 mm. If the anisotropy ratio r = 2, determine true strains in length, width and thickness directions. **[06]**

OR

Q-4(b) Draw a Keeler-Goodwin forming limit diagram and explain it in brief. **[06]**

Q-5 Answer following. (Any four) **[16]**

(i) What are lattice defects?
(ii) How do you represent strain hardening effect?
(iii) How do you determine yield strength of an alloy which exhibits a smooth curve in tensile test?
(iv) Define Formability and explain Olsen and Erichsen test for formability testing of sheet metal.
(v) How spiral submerged arc welded pipes are formed from the sheet metal/plate?
(vi) Draw a schematic diagram of Magnetic Pulse Forming and explain it in brief.

Q-6 Answer following questions. (Any one) **[08]**

(i) Explain the methodology of slip line field analysis in details.
(ii) Define computer simulation. Which softwares are used for analysis of machining and forming processes? What are the benefits of using softwares?

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